Predict the Introverts from the Extroverts

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Abstract

People's personalities often shine through in the way they interact with others, attend events, and share their lives. In this project, we explore the connection between social behaviors and personality types, focusing on how traits like extroversion and introversion shape and relate to our social patterns. Using the Extrovert vs. Introvert Personality Traits Dataset, which contains detailed measures of daily habits, event participation, and online activity, we aim to uncover patterns that link behavior to personality. Our main questions center on how these traits influence various aspects of social life, including friend circle size, online posting frequency, and the prediction of personality types based solely on recorded behaviors. By combining behavioral data with personality indicators, this project not only deepens our understanding of the link between social interactions and personalities but also explores the potential for predictive modeling.

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1. Introduction

People's personalities often shine through in the way they interact with others, attend events, and share their lives. In this project, we explore the connection between social behaviors and personality types, focusing on how traits like extroversion and introversion shape and relate to our social patterns. Using the **Extrovert vs. Introvert Personality Traits Dataset**, which contains detailed measures of daily habits, event participation, and online activity, we aim to uncover patterns that link behavior to personality.

Our main questions center on how these traits influence various aspects of social life. For instance, we ask: How do personality type, going-out frequency, and social event attendance affect the size of someone's friend circle? How do event participation and outdoor activity levels influence how often a person posts online? Finally, can we predict whether someone is

an introvert or extrovert based solely on their recorded behaviors?

By combining behavioral data with personality indicators, this project not only deepens our understanding of the link between social interactions and personalities but also explores the potential for predictive modeling. The findings could be relevant for fields like psychology and social network analysis, offering insights into how personality and behavior interact in the modern, interconnected world.

1.1 Dataset Description

1. Data files & provenance

1. **train.csv** – training set with both features and target; 18,524 rows, 9 columns; created by the course staff from the Kaggle dataset "Extrovert vs Introvert Behavior Data" (https://www.kaggle.com/competitions/playground-series-s5e7).
2. **test.csv** – test set (features only); 6,175 × 8; same source and cleaning procedure.

We removed duplicates and extreme outliers, balanced the introvert/extrovert ratio, and discretised a few numeric variables before releasing the three files.

2. Data structure

1. **Observational unit:** One respondent per row.
2. **Main columns (train.csv; test.csv excludes Personality):**
    - **id** (numerical)
    - **Time_spent_Alone:** Average hours alone per day (numerical, hours)
    - **Stage_fear:** Fear of speaking/performing in public (categorical)
    - **Social_event_attendance:** Number of social events attended per week (numerical; scaling to 0 - 10)
    - **Going_outside:** Number of times going outside per week (numerical; scaling to 0 - 7)
    - **Drained_after_socializing:** Feels exhausted after socialising (categorical)
    - **Friends_circle_size:** Count of close friends (numerical; scaling to 0 - 15)
    - **Post_frequency:** Social-media posts per week (numerical; scaling to 0 - 10)
    - **Personality:** Personality label Introvert/Extrovert (categorical)

3. Data preparation

1. **Missing values:** Before performing any analysis, we addressed missing values and standardized the dataset to ensure consistency across all models. We began by identifying columns with empty entries. Apart from the identifier (id) and the target variable (Personality), all other features contained missing values, which could introduce bias or distort model performance if left untreated. For categorical features, such as *Stage_fear* (≈10% missing) and *Drained_after_socializing* (≈6% missing), we replaced missing values using mode imputation. For numerical columns, where the missing proportion ranged from 5% to 8%, we applied median imputation. This approach ensured robustness against outliers while preserving the central tendency of each feature. Importantly, imputation was applied only to independent variables, as altering the target variable could affect the prediction validity.
2. **Data transformations:** We then transformed categorical variables into numerical form for modeling: binary "Yes/No" responses were encoded as 1 and 0, and Personality was encoded as 1 for extrovert and 0 for introvert. All numerical features were standardized using StandardScaler to ensure equal weighting across models.
3. **Model-ready split & evaluation:** For evaluation, we split the dataset using stratified sampling into 90% training and 10% hold-out sets. Within the training set, a 5/10-fold cross-validation procedure was employed. Regression tasks were evaluated using Mean Squared Error (MSE), while classification tasks used accuracy as the primary metric.
4. **Reproducibility:** All cleaning and feature steps wrapped in a single Pandas / scikit-learn Pipeline shared by the neural-network, random-forest, and linear/logistic models. We'll set the random seed to reproduce the result.

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2. Methodology and Experiments

2.1 Problem 1: Prediction of Stage Fright

Formulation

1. **Question:** Find 3 best independent variables in the dataset, then train a model to predict whether an individual has stage fright.
2. **Why it's interesting:** Stage fright is a common form of social anxiety. Understanding its relationship with fundamental personality traits and social energy patterns can offer insights into the psychological underpinnings of anxiety. If these factors are highly predictive, it could help identify individuals who might have stage fear and moreover, social anxiety.

Design

- **Data used:** A subset of the main dataset was created containing four columns: *Personality*, *Drained_after_socializing*, *Time_spent_Alone*, and *Stage_fear*. The final, cleaned dataset used for modeling is df_train01.

- **Data preparation:**

    1. *Missing value handling:* We first addressed missing values for the variables of interest. All rows with missing values in the target variable, *Stage_fear*, were removed. For the predictors, missing numerical data in *Time_spent_Alone* was replaced with the column's median, while missing categorical data in *Drained_after_socializing* was filled with the mode. This process ensures that our models are trained on a complete dataset.

    2. *Exploratory check:* To validate our choice of predictors, we computed the correlation coefficients between each predictor and the target variable, *Stage_fear*. The analysis revealed very strong relationships: corr(*Personality*, *Stage_fear*) ≈ -0.91, corr(*Drained_after_socializing*, *Stage_fear*) ≈ 0.88, and corr(*Time_spent_Alone*, *Stage_fear*) ≈ 0.80. These high correlation values strongly suggest that the selected features are highly informative for predicting stage fright and justify the use of classification models.

- **Models approach:** We fit and compared three supervised classification models to predict the stage fright:

    1. **K-Nearest Neighbors (KNN) Classifier** (non-parametric baseline). Hyperparameters tuned: n_neighbors (number of neighbors) and metric (distance calculation).

    2. **Logistic Regression** (linear, probabilistic model).

    3. **Random Forest Classifier** (nonlinear, tree-based ensemble).

- **Proposed Evaluation:** We will partition the data using 5-fold cross-validation to evaluate and compare the models. For the KNN classifier, we will use GridSearchCV to find the optimal combination of n_neighbors (from 1 to 29) and the distance metric (euclidean vs. manhattan). The Logistic Regression and Random Forest models will be evaluated using their default settings. The primary metrics for comparing model performance will be Accuracy, Precision, Recall, and F1-score. The model with the best overall performance on these metrics will be selected.

Implementation All preprocessing and modeling steps were encapsulated within scikit-learn Pipelines. A ColumnTransformer was configured to apply StandardScaler to the numerical

predictor (*Time_spent_Alone*) and OneHotEncoder to the categorical predictors (*Personality*, *Drained_after_socializing*). Following the design, we trained the three models. The KNN model's hyperparameters were tuned using GridSearchCV with cv=5 and scoring='accuracy'. The Logistic Regression and Random Forest models were evaluated using cross_val_score with 5 folds. To ensure reproducibility for the stochastic models (KNN and Random Forest), we set random_state=42. The final performance of each model was assessed using cross-validated predictions to calculate Accuracy, Precision, Recall, and F1-score.

Results The results show that all three models achieved exceptionally high performance, indicating a very strong predictive relationship between the chosen features and stage fright. The KNN Classifier is the best model. The high scores across the board suggest that personality, feelings of social drainage, and time spent alone are powerful determinants of whether an individual experiences stage fright.

Table 1: Performance Metrics of Different Models in Problem 1

Model	Accuracy	Precision	Recall	F1-score
KNN Classifier	0.993326	0.984154	0.988314	0.986230
Logistic Regression	0.992905	0.983168	0.987568	0.985363
Random Forest Classifier	0.992965	0.982695	0.988314	0.985496

Fig 1. Performance Metrics of Different Models in Problem 1

2.2 Problem 2: Social Activity and Post Frequency

Formulation

1. **Question:** Can we predict an individual's weekly social-media post frequency (*Post_frequency*) from two simple behavioral measures: *Social_event_attendance* and *Going_outside*?
2. **Why it's interesting:** Post frequency is a concrete indicator for online sociability. If in-person activity (how often one goes out and attends social events) is predictive of online posting, that reveals how offline social exposure maps onto online activity. This has applications in behavioral research and understanding the interplay between online/offline social life.

Design

- **Data used:**
    1. Subset of the main dataset with three columns: *Social_event_attendance*,

Going_outside, and *Post_frequency*.

2. After cleaning (see preprocessing below) the dataset used for modeling is df_train02.

- **Data preparation (question-specific):**

  1. *Missing value handling:* For this question we first performed targeted missing-value handling on the three variables of interest. We imputed missing entries in the predictors using robust central-tendency measures: *Going_outside* and *Social_event_attendance* were filled with their respective median values to reduce the influence of outliers. Rows with missing values in the target *Post_frequency* were removed because unlabeled examples cannot be used for supervised learning. After imputation and removal of any remaining nulls, we retained only the three columns required for modeling and assigned the cleaned table, which ensures the training set contains no empty entries for the predictors and only complete cases for the response.

  2. *Exploratory check:* We performed a simple exploratory check to confirm that modeling is appropriate. We computed Pearson correlation coefficients between each predictor and the target: corr(*Social_event_attendance*, *Post_frequency*) ≈ 0.547 and corr(*Going_outside*, *Post_frequency*) ≈ 0.553. Both correlations are around 0.5, indicating a moderately strong linear relationship with posting frequency and suggesting that the two behavioral features are informative predictors. These results motivate the choice to proceed with regression-based models and justify further model comparison and hyperparameter tuning.

- **Models:** We fit and compare three supervised regression models:

  1. **Ordinary least squares linear regression** (baseline).

  2. **Random Forest Regressor** (nonlinear, tree ensemble). Hyperparameters tuned: n_estimators (number of trees) and max_depth.

  3. **Feedforward neural network (MLPRegressor).** Hyperparameters tuned via grid search over hidden_layer_sizes and activation.

- **Proposed Evaluation:** We will begin by partitioning our dataset of 18,000 observations into a 90% training set and a 10% hold-out test set. Within the training set, we will perform 10-fold cross-validation to identify optimal hyperparameters for each modeling approach. Specifically, the neural network will be tuned over architectures of 2–5 hidden layers and 5–10 units per layer (as well as learning rate

and batch size); the random forest regressor will be evaluated across grids of tree counts (5,10) and maximum depths (None, 5, 10); the least squares linear regression model requires no tuning. Models with best 10-fold cross validation accuracy will be found, and the MSE of these models on the holdout set will be reported. Finally, we will use overall MSE as the main accuracy metric used to select the best model.

Implementation All preprocessing and modeling steps were implemented inside scikit-learn Pipelines to prevent data leakage and ensure that all transformations are learned only from training folds during cross-validation. Concretely, a ColumnTransformer was used to apply StandardScaler only to the two predictors (*Social_event_attendance* and *Going_outside*) while dropping any other columns; the response *Post_frequency* was never passed through the transformer. Median imputation for the two predictors was performed earlier in the data-preparation stage, so the pipeline is fed a cleaned df_train02 containing exactly the two features and the target. Placing scaling inside the pipeline guarantees that scale parameters are computed on each training fold rather than on the entire dataset.

Following the design, we trained three models (linear regression, random forest regressor, and MLP regressor) and evaluated them with 10-fold cross-validation. Hyperparameter tuning for the random forest (n_estimators, max_depth) and the neural network (hidden_layer_sizes, activation) was performed using GridSearchCV (with cv=10, and scoring = 'neg_mean_squared_error'). We set random_state=42 for stochastic components (random forest and MLP) to make results reproducible. For each model, negative MSE scores returned by scikit-learn were converted to positive MSE and RMSE for reporting, and final results were compiled into a summary table listing each model's cross-validated MSE and RMSE.

Results From the below results, we can see that all three models perform reasonably well, but the Random Forest and Neural Network outperform the Linear Regression model. The linear model's larger error suggests some non-linear relationship/ interaction between *Social_event_attendance* and *Going_outside* which Random Forest model and Neural Network model capture better.

Table 2: MSE/RMSE Results of Different Models in Problem 2

Model	MSE	RMSE
Linear Regression	5.005857	2.237377
Random Forest	3.987182	1.996793
Neural Network	3.982586	1.995642

Fig 2. MSE/RMSE Results of Different Models in Problem 2

2.3 Problem 3: Feature Selection for Personality Prediction

Formulation

1. **Question:** Using social behavior and personality-related features, which subset of attributes best predicts whether a person is an Introvert (0) or Extrovert (1), and which model achieves the highest classification performance?
2. **Why it's interesting:** Identifying a minimal yet effective set of behavioral indicators for personality classification has practical importance: fewer variables can reduce data collection costs, simplify models for deployment, and improve interpretability while maintaining predictive accuracy. This problem examines how feature selection combined with model comparison can optimize classification performance in personality prediction.

Design

- **Data used:**
    1. The main dataset with columns: *Time_spent_Alone*, *Stage_fear*, *Social_event_attendance*, *Going_outside*, *Drained_after_socializing*, *Friends_circle_size*, *Post_frequency*, and *Personality*.
    2. After cleaning and imputing missing values, the working dataset is stored as df_train_bestfeat.
- **Data preparation (question-specific):**
    1. *Missing value handling:* All predictor columns were converted to numeric, with unparseable entries set to NaN. Missing values in predictors were imputed using a constant value of 0 to handle even columns with all missing entries (e.g., *Stage_fear*). This ensured no feature subset would produce an empty matrix during scaling or modeling. Missing labels in *Personality* were removed. The cleaned dataset retained only complete target labels and fully imputed predictor columns.
    2. *Feature subset evaluation:* We enumerated all possible non-empty subsets of the seven predictor variables and evaluated each with three classifiers: Logistic Regression, Random Forest, and Gradient Boosting. Each model was embedded in a scikit-learn Pipeline with preprocessing (constant imputation, StandardScaler for linear models). For each subset, we split data into train/test (80/20 stratified) and computed ROC-AUC on the test set. All results were stored and sorted by AUC to find the top-performing model–feature subset combination.

- **Models:** We compared three supervised classification models:
    1. **Classifier 1:** Logistic Regression (linear baseline)
    2. **Classifier 2:** Random Forest Classifier (nonlinear tree ensemble)
    3. **Classifier 3:** Gradient Boosting Classifier (boosted decision trees)
- **Proposed Evaluation:** For each model, training and evaluation were conducted inside a Pipeline to avoid data leakage. Preprocessing included constant-value imputation for all predictors and feature scaling for linear models. Model selection was based on the highest ROC-AUC on the holdout test set.

Implementation

1. **Preprocessing:** SimpleImputer(strategy="constant", fill_value=0) to handle missing data; StandardScaler for Logistic Regression; tree-based models used raw imputed values.
2. **Search method:** Exhaustive search over all $2^7-1=127$ possible feature subsets. For each subset, fit all three models and record AUC.
3. **Selection:** Identify the single combination of model and subset with the maximum AUC.

Results The best-performing configuration was a Gradient Boosting Classifier using the following 5 features:

1. *Time_spent_Alone*
2. *Stage_fear*
3. *Social_event_attendance*
4. *Going_outside*
5. *Drained_after_socializing*

This combination achieved an AUC of 0.965 on the holdout set, outperforming all other model–feature combinations. The table below summarizes the top result versus the next best alternatives:

Table 3: Optimal Features & Test AUC Results of Different Models in Problem 3

Model	Features Used	Test AUC
Gradient Boosting	Time_spent_Alone, Stage_fear, Social_event_attendance, Going_outside, Drained_after_socializing	0.9650
Random Forest	Time_spent_Alone, Stage_fear, Social_event_attendance, Going_outside, Drained_after_socializing, Friends_circle_size	0.9621
Logistic Regression	Time_spent_Alone, Social_event_attendance, Going_outside, Drained_after_socializing	0.9574

Fig 3. Optimal Features & Test AUC Results of Different Models in Problem 3 Fig 4. ROC Curves of Best Model vs Alternatives

The selected 5 features cover multiple dimensions of social behavior: social frequency, comfort in public contexts, and post-socialization energy changes. These dimensions align with established psychological constructs distinguishing introversion and extraversion. Gradient Boosting's advantage suggests important nonlinearities and interactions (e.g., "high stage fear + high going outside" patterns) that linear models could not fully capture.

2.4 Problem 4: Full Feature Personality Prediction

Formulation

1. **Question:** Using social behavior and personality-related features (all attributes in the dataset), can we classify whether a person is an Introvert (0) or Extrovert (1)?
2. **Why it's interesting:** Personality classification from behavioral signals has applications in psychology, personalized services, and etc.. Here we examine whether simple behavioral measures strongly predict self-identified personality.

Design

- **Data used:**
    1. The whole main dataset with columns: *Time_spent_Alone*, *Stage_fear*, *Social_event_attendance*, *Going_outside*, *Drained_after_socializing*, *Friends_circle_size*, *Post_frequency* and *Personality*.
    2. After cleaning (see preprocessing below) the dataset used for modeling is df_train04.
- **Data preparation (question-specific):**
    1. *Missing value handling:* For this question we first performed targeted

missing-value handling on those eight variables of interest. We imputed missing entries in the predictors using robust central-tendency measures: numerical variables (*Time_spent_Alone*, *Friends_circle_size*, *Going_outside* and *Social_event_attendance*) were filled with their respective median values to reduce the influence of outliers; categorical variables (*Stage_fear* and *Drained_after_socializing*) were filled with their respective mode values to reduce the influence of outliers. Rows with missing values in the target *Personality* were removed because unlabeled examples cannot be used for supervised learning. After imputation and removal of any remaining nulls, we retained only the eight columns required for modeling and assigned the cleaned table, which ensures the training set contains no empty entries for the predictors and only complete cases for the response.

2. *Exploratory check:* We performed a simple exploratory check to confirm that modeling is appropriate. We computed Pearson correlation coefficients between each predictor and the target (see below). All correlations (in absolute value) are above 0.6, indicating a moderately strong linear relationship with posting frequency and suggesting that the behavioral features above are informative predictors. These results motivate the choice to proceed with classifier-based models and justify further model comparison and hyperparameter tuning.

Fig 5. Absolute Correlation Between Each Features and Personality

- **Models:** We fit and compare three supervised regression models: a. **Classifier 1:** Full-connected neural network b. **Classifier 2:** A random forest classifier c. **Classifier 3:** The K Nearest Neighbors Classifier

- **Proposed Evaluation:** All modeling steps were implemented inside scikit-learn Pipelines to guarantee that preprocessing is learned only on training folds and to avoid leakage: a ColumnTransformer applied StandardScaler to numeric features and OneHotEncoder to categorical features, then the classifier sat at the pipeline's end. Hyperparameter grids were explored with GridSearchCV (stratified cv=5): KNN was searched over n_neighbors from 1 to 29 and distance metrics ['euclidean','manhattan']; Random Forest was tuned over n_estimators ∈ {100, 200, 500} and max_depth ∈ {None, 10, 20, 30}; the MLP tested a range of small multi-layer architectures (single and multi-layer tuples up to five hidden layers, e.g. (5,5), (10,10,5,5,10)) with activations ['relu','tanh'] and max_iter=1000 to ensure convergence. Models with best 5-fold cross validation accuracy will be found, and the accuracy, precision, recall, and F1 of these models on the holdout set will be reported. Finally, we will use overall accuracy as the main metric used to select the best model.

Implementation All modeling steps were implemented as scikit-learn Pipelines that combine a ColumnTransformer (applying StandardScaler to numeric features and OneHotEncoder to categorical features) with the classifier, so that scaling and encoding are performed inside each CV fold and data leakage is avoided.

Hyperparameter tuning was performed with grid search: for K-Nearest Neighbors we searched n_neighbors from 1 to 29 and distance metrics ['euclidean','manhattan']; for Random Forest we searched n_estimators in {100, 200, 500} and max_depth in {None, 10, 20, 30}; and for the MLP we tested a range of small multi-layer architectures (single and multi-layer tuples up to five hidden layers, e.g. (5,5) or (10,10,5,5,10)) with activations ['relu','tanh'] and max_iter=1000 to ensure convergence. Additionally, We set random_state=42 for stochastic components for our models to make results reproducible. Final evaluation used cross_val_predict with cv=5 to produce out-of-fold predictions, and accuracy, precision, recall, and F1 were computed on those predictions so the reported metrics are unbiased by training.

Results From the results below, we can see that all three models perform exceptionally well on this classification task. The Neural Network Classifier slightly outperforms the other two models, achieving the highest scores in Accuracy, Precision, and F1-score. However, the performance differences between the three are minimal, indicating that the K-Nearest Neighbors and Random Forest models also effectively capture the relationships between a person's social behaviors and their personality type (introvert vs. extrovert).

Table 4: Performance Metrics of Different Models in Problem 4

Model	Accuracy	Precision	Recall	F1-score
KNN Classifier	0.968851	0.977302	0.980656	0.978976
Logistic Regression	0.968527	0.977015	0.980510	0.978759
Random Forest Classifier	0.969121	0.977728	0.980583	0.979153

Fig 6. Performance Metrics of Different Models in Problem 4

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3. Discussion

Across all four problems a single, unifying theme emerges: simple, well-chosen measures of social activity and affect (e.g. time spent alone, stage fear, social event attendance, going-out frequency, drained-after-socializing) carry strong, stable signal about both behavioral outcomes (post frequency) and latent traits (introversion vs. extraversion). The models—linear, tree-based, nearest-neighbour and neural nets—agree on the substantive story: these

features together explain much of the variation in stage fright, posting behavior, and personality label. That agreement is informative:

1. **Strong predictive signal, modest model-dependence.** High scores in Problems 1 and 4 (accuracies ≈ 0.97–0.99) and the AUC ≈ 0.965 (Problem 3) show the problem is well-posed for statistical prediction. Different algorithms (KNN, RF, Gradient Boosting, NN) reach similarly excellent performance, which suggests the signal-to-noise ratio in the chosen features is high — not an artifact of a single modeling choice.
2. **Nonlinearity and interactions matter.** The regression comparison in Problem 2 (Linear vs Random Forest vs Neural Net) indicates that relationships between social attendance, going-out frequency and post frequency are not strictly linear: tree ensembles and NNs reduced MSE notably. Similarly, Gradient Boosting being best for the five-feature introversion model (Problem 3) points to useful nonlinear interactions (e.g. stage fear moderating the effect of going out).
3. **Parsimonious feature sets suffice.** Problem 1 showed that just three features can almost perfectly predict stage fright; Problem 3 found a compact 5-feature set that yields near-optimal AUC. This implies that a small, interpretable battery of measures can be practically useful for prediction or screening.
4. **Ensemble / flexible learners excel, but simple models compete.** While GBM and NN capture subtle interactions, KNN and logistic models produced nearly as good results. In applied settings this suggests a trade-off: prefer simpler models for interpretability and deployment, but use flexible models when maximizing predictive accuracy is critical.

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4. Conclusion

This project demonstrates that a few key measures of social behavior and emotional response—such as time spent alone, fear of public speaking, frequency of attending social events, how often one goes out, and feelings of being drained after socializing—are powerful and reliable predictors of both specific behaviors and underlying personality traits.

Across four distinct analytical problems, our models consistently showed that these features contain a strong signal for predicting outcomes like social media post frequency and, most notably, for classifying individuals as introverts or extroverts. The high accuracy, precision, and AUC scores achieved by various models underscore the robustness of these findings. While different algorithms were employed, their similarly high performance suggests that the predictive power comes from the data itself, rather than being an artifact of a specific

modeling technique.

A key takeaway is that comprehensive data collection is not always necessary for accurate personality prediction. We found that a concise set of just five features was nearly as effective as the full set, suggesting that a small, well-chosen battery of questions can be highly effective for practical applications like screening or preliminary assessment. While complex models can offer a slight advantage in predictive accuracy, simpler, more interpretable models also performed exceptionally well, presenting a viable trade-off for applications where transparency and ease of deployment are priorities.